

Team SuperTronics

IOT BASED IV-BAG MONITORING AND ALERT SYSTEM

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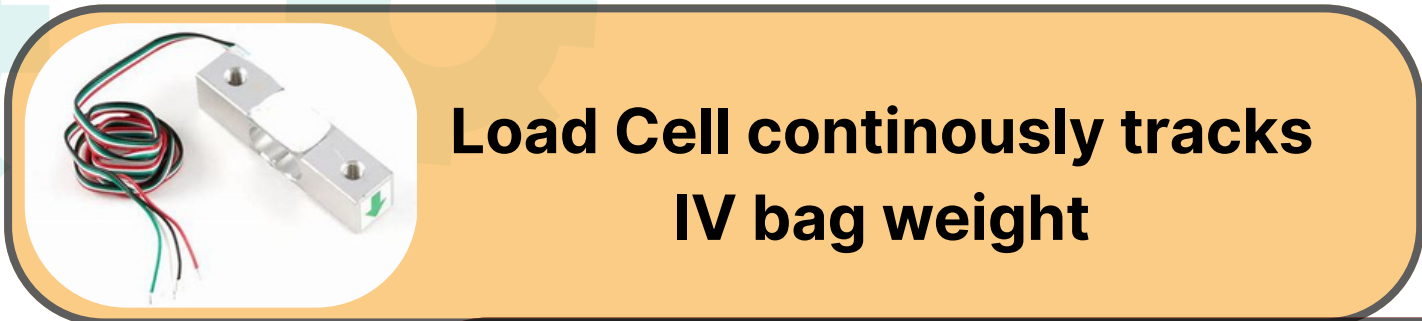
The Problem with Monitoring IV Manually

- Human Errors - Nurses may forget to check bags regularly
- Delayed Response - Critical drops in fluid levels may go unnoticed
- Patient Safety Risks - Backflow, Embolisms, Complications
- Insuffiecient Use of Staff - Time wasted on manual checks

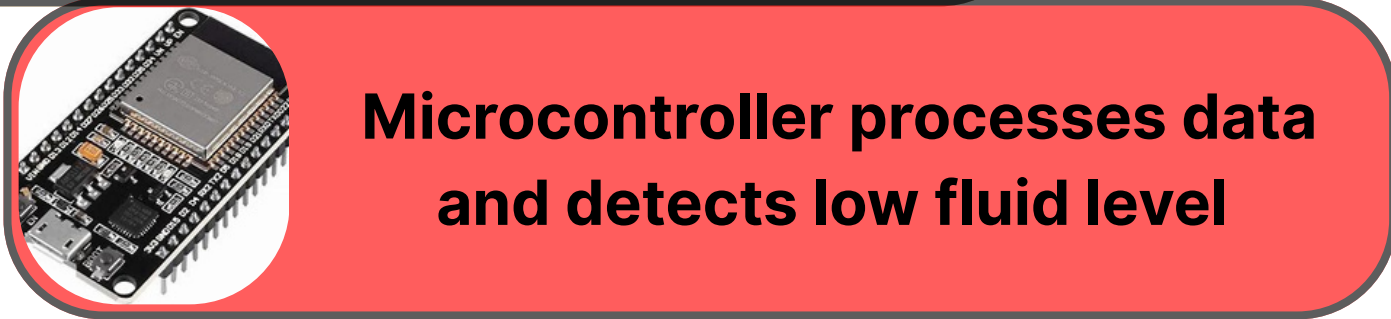




A Smarter & Safer IV Monitoring System



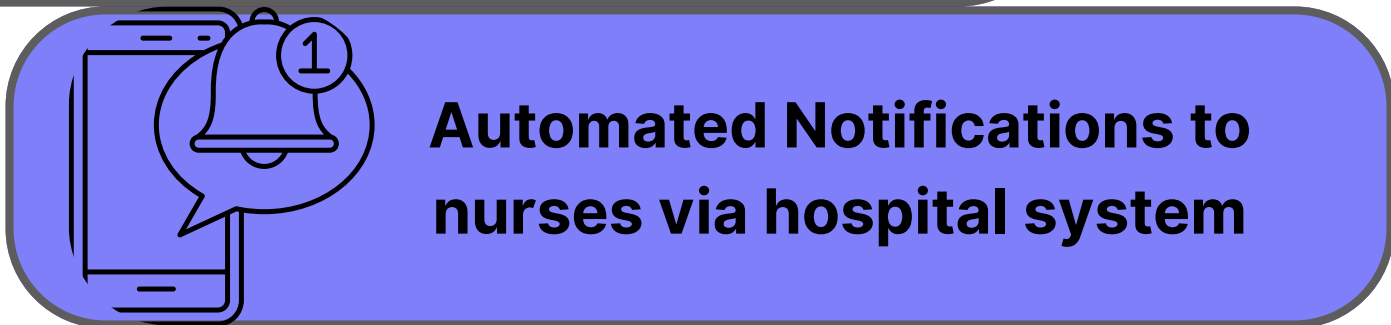
Load Cell continuously tracks IV bag weight



Microcontroller processes data and detects low fluid level



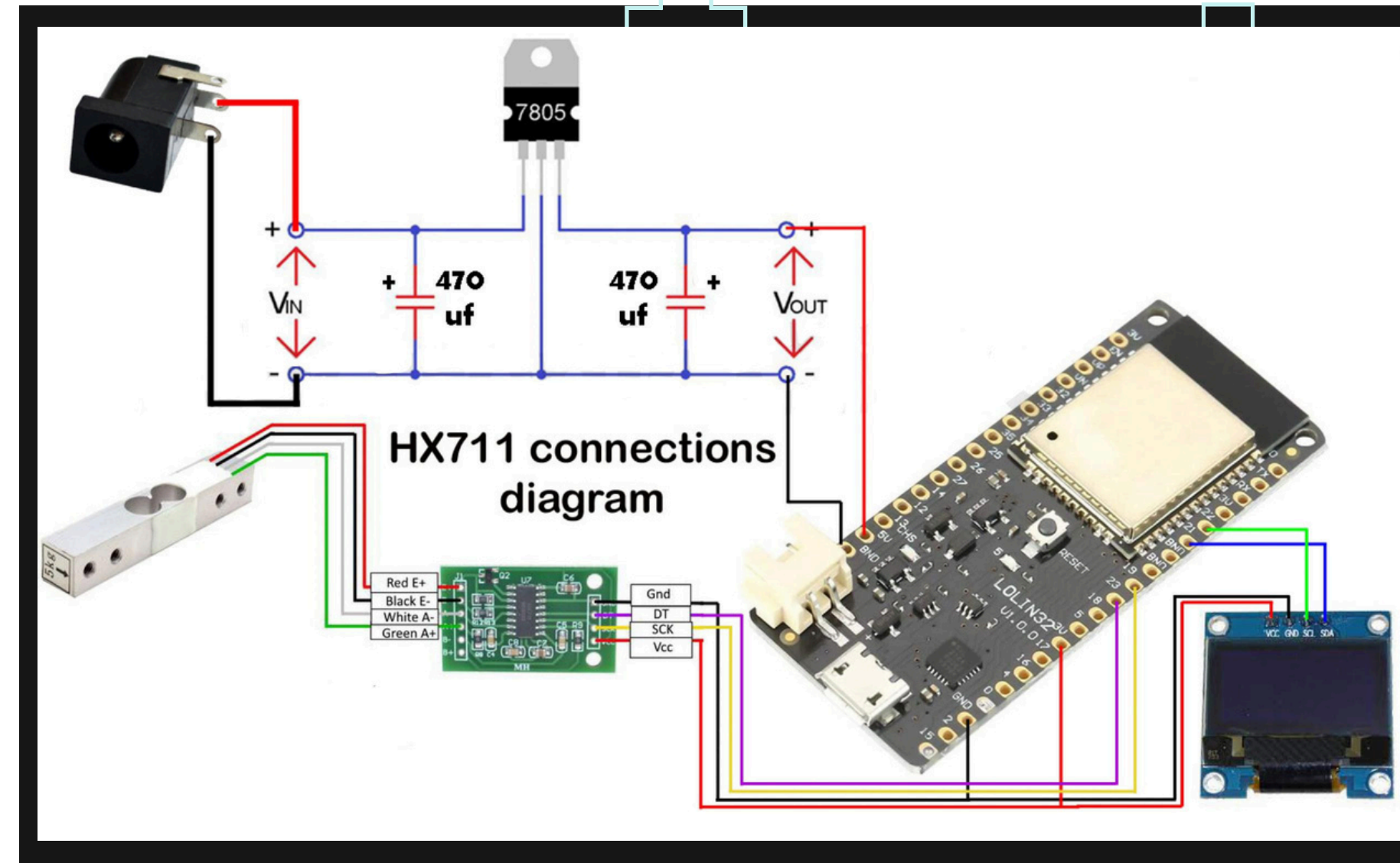
Setting alert threshold manually



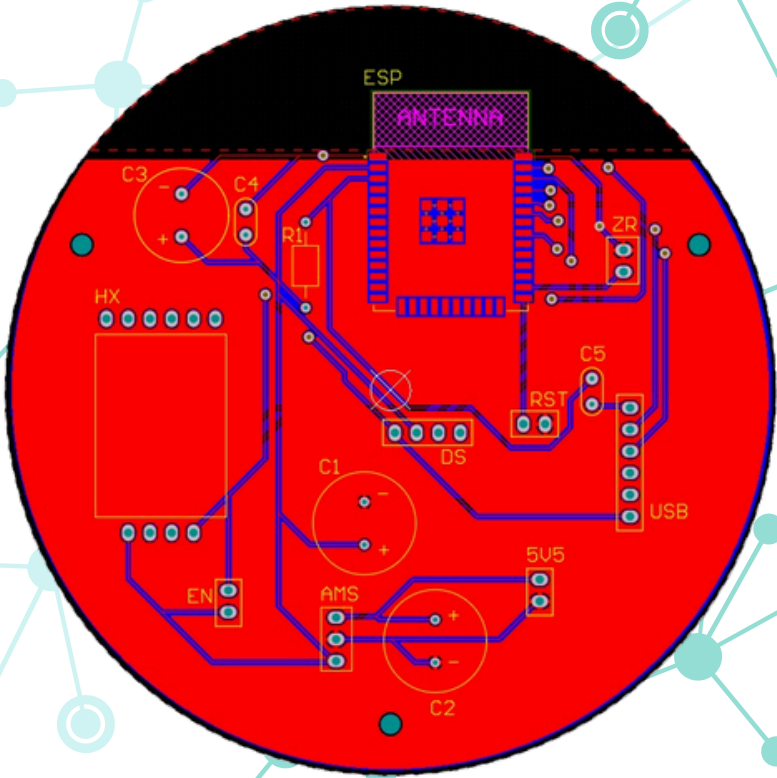
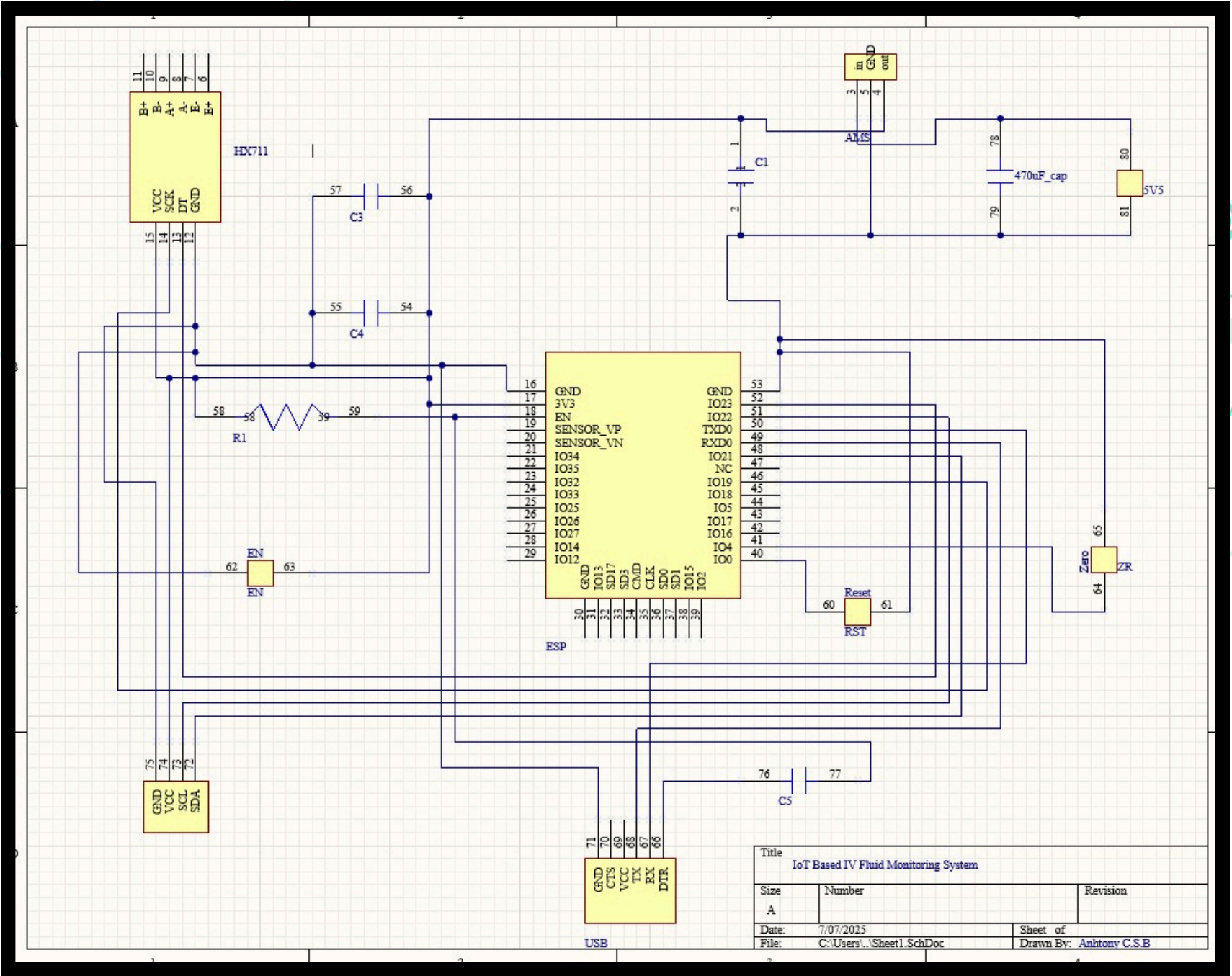
Automated Notifications to nurses via hospital system



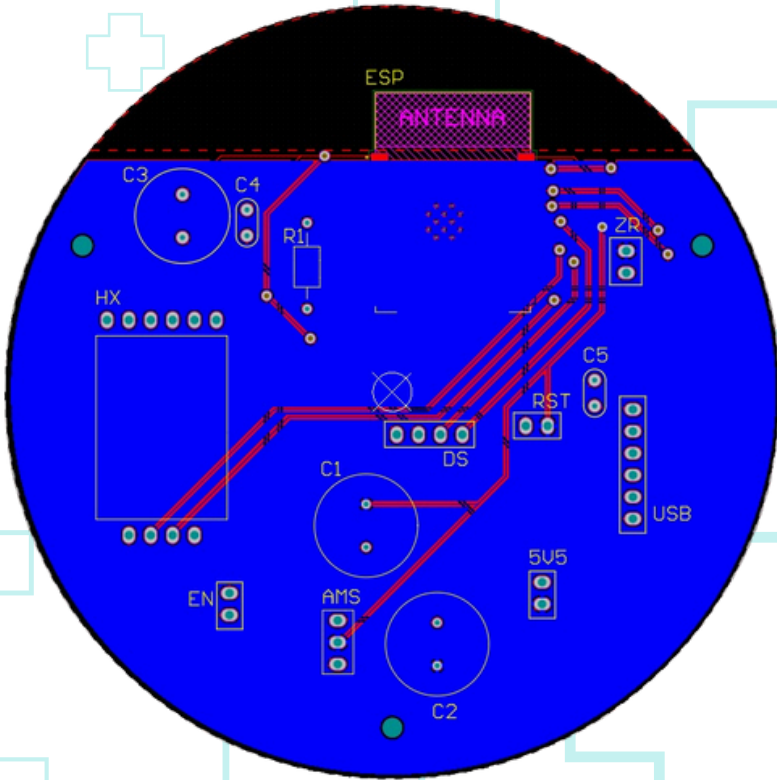
Remote Monitoring by doctors and nurses



PCB Schematic



Top
Layer



Bottom
Layer

Cost Analysis

Component	Cost (LKR)
PCB	400
ESP32	1400
Load Cell	350
Load Cell Bracket	1300
Enclosure	6100
HX711 Module	150
Display (OLED)	800
Other Components	500
Total	11000

How to use

1. Hang the device and set the initial value to Zero using the tare button on top.
2. Hang the IV bag and set the recorded weight as the initial value in the web interface.
3. Set the desired threshold value as a percentage of the initial weight through the web interface.

This is not merely just a project
It's a leap towards safer and smarter healthcare